A Comparative Study of Supervised and Self-Supervised Learning on BCI Competition IV-2a Spectrogram Data

1st Kazi Maisha Mehzabin
Dept. of Computer Science and Engineering
East West University
ID: 2022-2-60-108
Dhaka, Bangladesh
meh.zabin@email.com

2nd Istiaque Ahmed
Dept. of Computer Science and Engineering
East West University
ID: 2022-2-60-096
Dhaka, Bangladesh
istiaque.ahamed@email.com

3rd Saba Tasnim Khan
Dept. of Computer Science and Engineering
East West University
ID: 2022-3-60-049
Dhaka, Bangladesh
saba@email.com

4th Rubaiya Zaman Mysha
Dept. of Computer Science and Engineering
East West University
ID: 2022-1-60-174
Dhaka, Bangladesh
rubaiya@email.com

***Abstract*—This report provides a summary of the final results gained from the given Kaggle notebooks and submitted reports for Tasks 1 and 2 of the BCI Competition IV-2a spectrogram data set. The research considers five supervised baselines and three self-supervised learning approaches: SimCLR, BYOL, and Barlow Twins. The supervised models were tested using train:test splits, whereas the SSL analysis contains the results of the 80:20 train:test split with embeddings, k-NN classification accuracy, label efficiency analysis, ablation, and statistical tests computed using the notebooks. In terms of supervised baselines, the best result was observed when testing ResNet50 with an 80:20 train:test split, which gave an accuracy score of 0.2979. In terms of the SSL models, BYOL linear probe gave an accuracy of 0.2824, while Barlow Twins fine-tuning provided the best end-to-end SSL accuracy result of 0.2689. This report is limited to only the results and figures available in the uploaded notebooks and previous reports.**

***Index Terms*—BCI Competition IV-2a, Spectrogram classification, Supervised learning, Self-supervised learning, SimCLR, BYOL, Barlow Twins**

I. Introduction

The project study involves recognition of multiple classes through spectrograms created using the BCI Competition IV Dataset 2a dataset. These classes include Feet, Left Hand, Right Hand, and Tongue. The final report synthesizes three documents that were previously generated as part of the project process, namely the related-work information from Task 1, the split-wise supervised baseline approach from Task 2, and the final notebook containing the output from the three self-supervised learning approaches.

The objective of the final paper is to provide a comparison rather than any speculation. This document follows the structure expected by Task 8, which includes related work, methods, experiment, results, ablation with statistical analysis, discussion, limitations, and conclusions. If an item that was expected by the task could not be printed in the notebook, it is not fabricated in this report.

II. Related Work

Task 1 outlined the benchmark data set and three exemplary previous approaches. A compact version of the submitted related-work summary is presented in Table I.

III. Methods

A. Supervised Baselines

The supervised experiment includes five classifiers on images: EfficientNet-B0, MobileNetV2, ResNet50, ViT-Tiny, and ConvNeXt-Tiny. According to the revised report of Task 2, the number of images in the data set is 2,583, and the input size for all runs is fixed to 224 $\times$ 224. The reported train:test split ratios are 10:90, 20:80, 30:70, 40:60, 50:50, 60:40, 70:30, 80:20, and 90:10.

B. Self-Supervised Learning Methods

The notebook applies three self-supervised learning algorithms: SimCLR, BYOL, and Barlow Twins. In the notebook summary, the SSL input size is 96 $\times$ 96, while downstream evaluation is performed on 224 $\times$ 224. The three SSL approaches are evaluated in terms of linear probing and fine-tuning for downstream classification.

TABLE I
RELATED-WORK SUMMARY REPRODUCED FROM THE SUBMITTED TASK 1 MATERIAL.

Work	Method	Reported Result	Note
BCI Competition IV Dataset 2a (official description)	Dataset reference	N/A	Benchmark specification used as the standard dataset
EEG-TCNet: An accurate temporal convolutional method for embedded motor imagery BCIs	EEG-TCNet	77.35% average; 83.84% subject-specific tuned	Low computational cost and strong benchmark performance
A Methodology Based on Filter Bank Common Spatial Pattern for Multiclass Motor Imagery BCI	MO-FBCSP	Competitive multiclass performance	Interpretable classical baseline with spatial-frequency features
EEG GENet: A feature-level graph embedding-based motor imagery classification model for EEG signals	EEG GENet	79.57% on BCICIV-2a	Graph-based channel modeling with good benchmark accuracy

C. Derived Analyses from the Notebooks

Aside from downstream accuracy measures, the notebooks also provide other analyses required by Task 8, such as embeddings by PCA, t-SNE, and UMAP with silhouette scores, k-NN classification accuracy based on the learned embedding space, label-efficiency analysis, ablation studies, pairwise statistical tests, and SSL ratio analysis.

IV. EXPERIMENTAL SETUP

TABLE II
EXPERIMENTAL SETTING COMPILED FROM TASK 2 AND THE FINAL SSL NOTEBOOK.

Item	Value
Dataset	BCI Competition IV Dataset 2a spectrogram images
Total images	2,583
Classes	Feet, Left Hand, Right Hand, Tongue
Supervised input size	224 × 224
SSL input size	96 × 96 (notebook summary)
Supervised models	EfficientNet-B0, MobileNetV2, ResNet50, ViT-Tiny, ConvNeXt-Tiny
SSL methods	SimCLR, BYOL, Barlow Twins
Reported supervised splits	10:90, 20:80, 30:70, 40:60, 50:50, 60:40, 70:30, 80:20, 90:10

TABLE III
REPORTED SAMPLE COUNTS FOR THE SUPERVISED TRAIN:VALIDATION:TEST SPLITS.

Split	Train	Validation	Test
10:90	232	26	2325
20:80	464	52	2067
30:70	696	78	1809
40:60	929	104	1550
50:50	1161	130	1292
60:40	1394	155	1034
70:30	1627	181	775
80:20	1859	207	517
90:10	2091	233	259

V. RESULTS

A. Supervised Baseline Overview by Split

The supervised split-based analysis from the modified Task 2 report reveals one best model for each of the reported train:test ratio pairs. The split with an 80:20 ratio yields the highest overall performance under supervised training, where ResNet50 achieves 0.2979 accuracy, 0.2784 macro F1-score, and 0.5418 ROC-AUC score.

TABLE IV
BEST SUPERVISED MODEL FOR EACH SPLIT, REPRODUCED FROM THE SUBMITTED TASK 2 REPORT.

Split	Best Model	Accuracy	Macro F1	ROC-AUC
10:90	ViT-Tiny	0.2624	0.2217	0.5023
20:80	ResNet50	0.2675	0.2433	0.5169
30:70	ConvNeXt-Tiny	0.2631	0.2025	0.5137
40:60	ResNet50	0.2561	0.2182	0.4980
50:50	ResNet50	0.2771	0.2701	0.5273
60:40	ViT-Tiny	0.2660	0.1764	0.5208
70:30	EfficientNet-B0	0.2723	0.2692	0.5173
80:20	ResNet50	0.2979	0.2784	0.5418
90:10	ConvNeXt-Tiny	0.2857	0.1870	0.5294

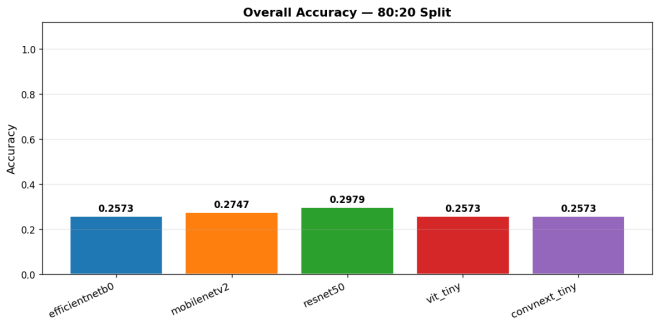

Fig. 1. Overall accuracy comparison for the 80:20 supervised split.

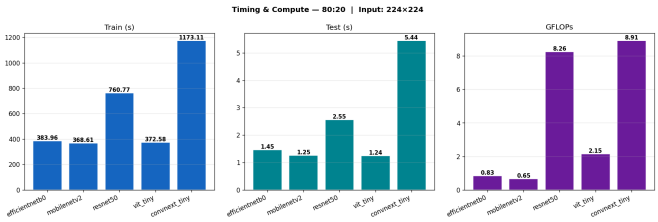

Fig. 2. Training time, testing time, and GFLOPs for the 80:20 supervised split.

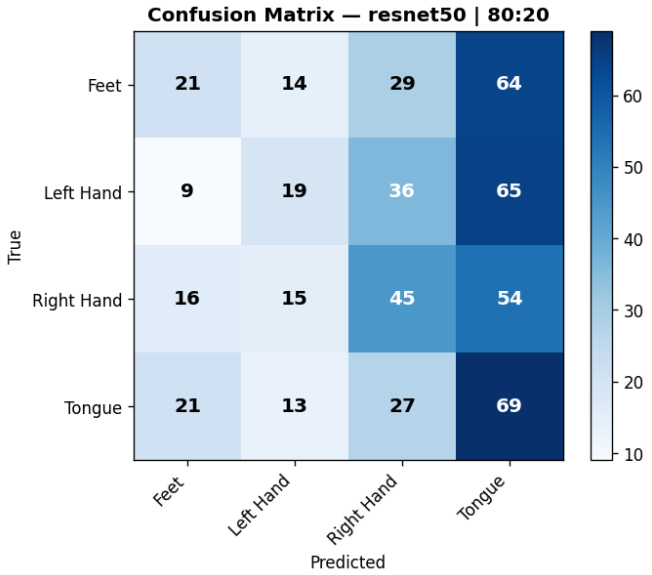

Fig. 3. Confusion matrix of the best 80:20 supervised baseline (ResNet50).

For the best supervised 80:20 model, the Task 2 report identifies that the class where the errors are most concentrated is Tongue, with instances where there were false identifications such as Left Hand to Tongue, Feet to Tongue, and Right Hand to Tongue.

B. Final 80:20 Comparison of Supervised and SSL Methods

The final SSL notebook contains a consolidated table for comparing the results of the models for the 80:20 scenario. The optimal frozen-features SSL model is BYOL linear probe, which had an accuracy of 0.2824, whereas the optimal end-to-end SSL model is Barlow Twins fine-tuning, with an accuracy of 0.2689.

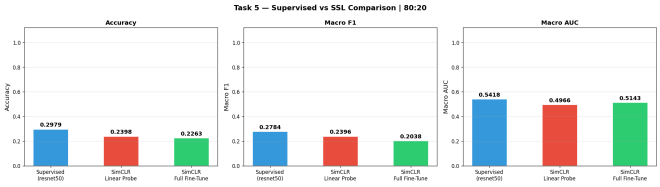

Fig. 4. Notebook-generated comparison of the best supervised baseline versus SimCLR linear probe and SimCLR full fine-tuning on the 80:20 split.

C. Embedding, k-NN, and Label-Efficiency Analysis

In the notebook, embedding is analyzed for all SSL encoding schemes using PCA50, t-SNE, and UMAP. All the values of the silhouette score obtained are negative.

VI. Ablations with Statistics

In the ablation study, the SimCLR temperature and Barlow Twins projection-head depth are varied. In the notebook, the optimal temperature for SimCLR is 0.1, whereas the optimal Barlow Twins depth is 3.

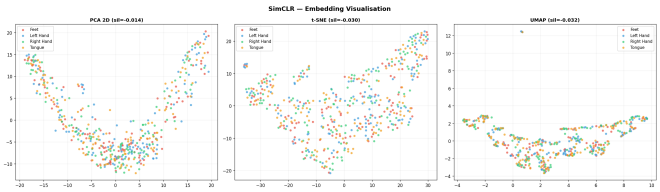

Fig. 5. Notebook-generated embedding analysis for SimCLR on the 80:20 setting.

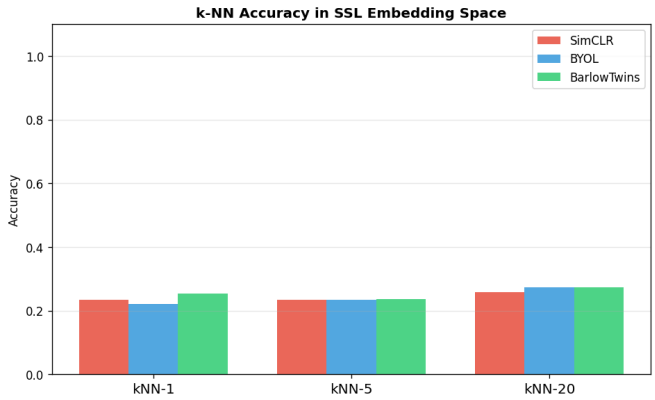

Fig. 6. Notebook-generated k-NN accuracy summary for SSL embeddings.

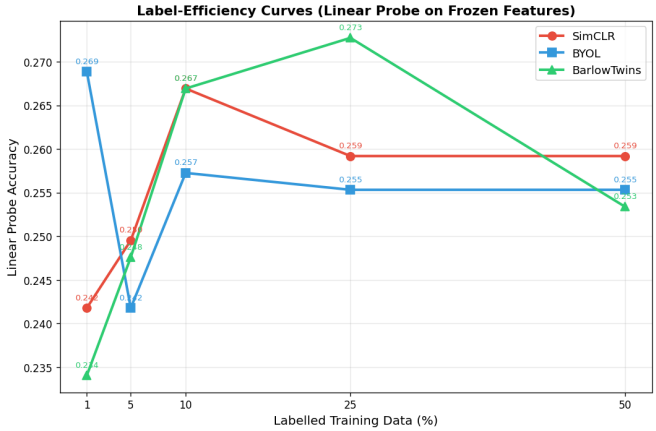

Fig. 7. Label-efficiency curves generated by the notebook for the three SSL methods.

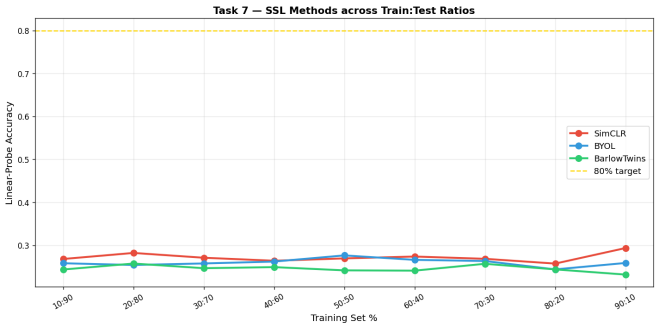

Fig. 8. Notebook-generated ablation plots for SimCLR temperature and Barlow Twins projection-head depth.

TABLE V
FINAL 80:20 COMPARISON TABLE COMPILED FROM THE TASK 8 NOTEBOOK SUMMARY.

Method	Accuracy	Macro F1	Macro AUC	Labels Used
Supervised - ResNet50	0.2979	0.2784	0.5418	All
BYOL - Linear Probe	0.2824	0.2722	0.5353	Head only
Barlow Twins - Fine-Tune	0.2689	0.2533	0.5145	All (end-to-end)
Barlow Twins - Linear Probe	0.2611	0.2400	0.5140	Head only
BYOL - Fine-Tune	0.2611	0.1893	0.5030	All (end-to-end)
SimCLR - Linear Probe	0.2398	0.2396	0.4966	Head only
SimCLR - Fine-Tune	0.2263	0.2038	0.5143	All (end-to-end)

TABLE VI
SILHOUETTE SCORES FROM THE NOTEBOOK EMBEDDING ANALYSIS.

Method	PCA50	t-SNE	UMAP
SimCLR	-0.0142	-0.0300	-0.0320
BYOL	-0.1879	-0.0348	-0.0449
Barlow Twins	-0.0298	-0.0360	-0.0374

TABLE VII
K-NN ACCURACY IN SSL EMBEDDING SPACE.

Method	k=1	k=5	k=20
SimCLR	0.2340	0.2340	0.2592
BYOL	0.2205	0.2340	0.2747
Barlow Twins	0.2534	0.2360	0.2747

In the case of the five-way linear probe experiment, the mean accuracies for SimCLR, BYOL, and Barlow Twins were recorded as 0.2541, 0.2435, and 0.2415, respectively. The result from the Friedman test shows that the statistic equals 3.8947 and the p-value equals 0.1426, which is interpreted by the notebook as indicating no statistically significant difference at the 0.05 significance level.

VII. DISCUSSION

Based on the analysis of the provided materials in this report, the supervised ResNet50 model on the 80:20 split achieves the highest accuracy. Within self-supervised learning, in terms of frozen-feature representation via linear probing, BYOL performs better than the other methods, and Barlow Twins demonstrates better accuracy during the full fine-tuning procedure. It appears from the results of embedding and label-efficiency analyses that the performance of SSL methods varies according to the evaluation approach used; specifically, BYOL is better at 1%, whereas Barlow Twins outperforms the others at 25%.

Moreover, the analysis of the confusion matrix in Task 2 reveals inter-class confusion among supervised methods. This corresponds to the moderate accuracy achieved by both approaches and their low silhouette scores in the SSL embedding analysis. Therefore, the current research should be considered comparative rather than absolute, as none of the methods has shown high accuracy in multiclass classification of spectrograms.

TABLE VIII
LABEL-EFFICIENCY ACCURACY VALUES REPORTED BY THE NOTEBOOK.

Label Fraction	SimCLR	BYOL	Barlow Twins
1%	0.2418	0.2689	0.2340
5%	0.2495	0.2418	0.2476
10%	0.2669	0.2573	0.2669
25%	0.2592	0.2553	0.2727
50%	0.2592	0.2553	0.2534

TABLE IX
SIMCLR TEMPERATURE ABLATION.

SimCLR temperature	Linear-probe accuracy
0.1	0.2689
0.3	0.2553
0.5	0.2476
0.7	0.2360
1.0	0.2456

TABLE X
BARLOW TWINS PROJECTION-HEAD-DEPTH ABLATION.

BT projection-head depth	Linear-probe accuracy
1	0.2456
2	0.2553
3	0.2611

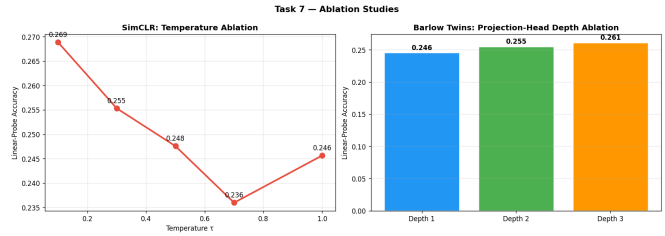

Fig. 9. SSL linear-probe accuracy across multiple train:test ratios reported by the notebook.

VIII. LIMITATIONS AND ETHICS

This report focuses on the output from the existing notebooks and previous reports. For instance, the Task 2 report clearly specifies that no ROC curves were created in the notebooks, which is why ROC-AUC scores are mentioned here, but no plots were fabricated. This rule was followed for each section and figure in this report.

A second limitation is that the performance levels are relatively low in absolute terms. Due to significant class confusion, the results should not be viewed as an already deployed product but rather as comparative results between baselines.

IX. Conclusion

The final project data show that the supervised ResNet50 model with an 80:20 train:test ratio is the top reference baseline with 0.2979 accuracy, 0.2784 macro F1, and 0.5418 macro AUC. Among the three self-supervised learning techniques, BYOL linear probe is the best-performing model when using frozen features, whereas Barlow Twins fine-tuning is the most effective approach within end-to-end self-supervised learning. The additional analysis includes embeddings, k-NN, label efficiency, ablation, and ratio analyses, but it does not surpass the supervised model in the submitted results.

Author Contributions

Kazi Maisha Mehzabin (2022-2-60-108): 37%
Istiaque Ahmed (2022-2-60-096): 39%
Saba Tasnim Khan (2022-3-60-049): 12%
Rubaiya Zaman Mysha (2022-1-60-174): 12%

References

[1] "BCI Competition IV Dataset 2a Official Description," *BCI Competition IV*. [Online]. Available: https://www.bbci.de/competition/iv/

[2] R. Mane, T. Chouhan, and C. Guan, "EEG-TCNet: An Accurate Temporal Convolutional Network for Embedded Motor-Imagery Brain-Machine Interfaces," *arXiv preprint arXiv:2006.00622*, 2020. [Online]. Available: https://arxiv.org/abs/2006.00622

[3] J. Jin, Z. Yu, I. Ullah, K. Wang, and T. Xu, "A Method Based on Filter Bank Common Spatial Pattern for Multiclass Motor Imagery BCI." [Online]. Available: https://www.researchgate.net/publication/337078235

[4] X. Zhang, X. Chen, L. Wang, J. Gao, and B. Li, "EEG GENet: A Feature-Level Graph Embedding Method for Motor Imagery Classification Based on EEG Signals." [Online]. Available: https://www.researchgate.net/publication/363079313